3.2 Factoring Special Products

Name _____ Period ____ Date _____

Algebra II 3.2 (30-minute plan) · Student sheet

I can…

- factor the difference of two squares.
- factor perfect square trinomials.
- factor completely and solve equations using special product patterns.

Vocabulary · fill in the missing word

Difference of two squares — $a^2 - b^2 =$ _____

Perfect square trinomial — $a^2 + 2ab + b^2 =$ _____ and $a^2 - 2ab + b^2 = (a - b)^2$

Factored completely — no factor can be factored _____

Part 1 · Notes and examples (fill in each note, then work the example)

Difference of two squares: $a^2 - b^2 =$ _____

Example 1

Factor each expression.

a. $x^2 - 49$ b. $9y^2 - 16$

Perfect square trinomials: $a^2 + 2ab + b^2 =$ _____ $a^2 - 2ab + b^2 =$ _____

Example 2

Factor each expression.

a. $x^2 - 10x + 25$ b. $4x^2 + 12x + 9$

Sum of two squares: A sum of two squares, $a^2 + b^2$, _____.

Example 3

Factor completely: $3x^3 - 12x$

Part 2 · Partner practice

Partner practice

Factor $25x^2 - 1$.